



MG2008 – DATA ANALYSIS FOR BUSINESS-1
NAVIGATING THE AI & ACADEMIA PARADOX
THE TRENDS, ETHICS & IMPACTS
FINAL PROJECT REPORT

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Abstract

This research explores the evolving impact of Artificial Intelligence (AI) tools on student performance, academic integrity, and future workplace preparedness. In an age where AI is increasingly embedded into educational practices, this study aims to analyse how students are adopting these tools, how it affects their learning outcomes, and whether it fosters or hinders real-world readiness.

Primary data was collected from university students via a structured Google Form, covering variables such as AI usage frequency, purpose of use, CGPA, ethical perceptions, and demographic attributes. The analysis applied both descriptive and inferential statistical techniques—including frequency distributions, measures of central tendency and dispersion, z-scores, box plots, correlation, and simple linear regression.

Key findings indicate that frequent AI usage tends to correlate positively with higher self-reported GPAs, although ethical concerns remain a significant issue. Usage behaviour varied across demographic segments, with notable trends in how students integrate AI into tasks like brainstorming and assignment writing.

The study concludes that while AI tools hold potential to enhance academic support, their role must be critically evaluated within ethical boundaries to ensure academic integrity and genuine skill development. These insights are crucial for educators, policymakers, and institutions aiming to balance AI integration with academic standards.

Acknowledgement

First and foremost, I express my deepest gratitude to the Almighty for granting me the strength, patience, and clarity to complete this research successfully.

I would like to extend sincere appreciation to my course instructor and research supervisor *Miss Iman Illyas* for their consistent guidance, encouragement, and valuable feedback throughout the process. Their academic insight and support were instrumental in shaping this research.

I am also thankful to all the students who participated in the survey and contributed their honest responses, making this study possible with real and relevant insights.

Lastly, I would like to acknowledge *my group members* who were part of this project and the moral support from my peers and family, who motivated me during challenging moments of this journey. Although circumstances required me to take full charge of the execution and completion of this research, the overall experience contributed significantly to my growth as a student researcher.

Introduction & Literature Review

Stage 01 in the research

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NAVIGATING THE AI & ACADEMIA PARADOX

Introduction & Literature Review:

Background of the Study:

The integration of Artificial Intelligence (AI) tools into educational settings has reshaped how students learn, access information, and complete assignments. With tools like ChatGPT, Grammarly, and Google Gemini gaining rapid popularity, students now have access to intelligent assistants that can generate content, provide feedback, and simulate real-time tutoring. While such advancements offer new possibilities for enhancing productivity and understanding, they also raise important questions about the integrity of student work, the depth of learning, and the authenticity of academic outputs.

As AI becomes a prominent part of modern education, it is essential to understand how students interact with these tools and whether their use leads to genuine intellectual growth or fosters dependency. This study investigates these dynamics with a focus on academic integrity, academic performance, and students' preparedness for the workplace in an AI-powered educational landscape.

Relevance of AI in Academia:

AI technologies are no longer confined to labs or tech industries — they are embedded in classrooms, assignments, research work, and even examinations. Tools like Turnitin have introduced AI-detection mechanisms, while universities globally are revisiting their academic honesty policies in light of emerging technologies.

For students, AI offers convenience, speed, and tailored feedback — often acting as a silent co-pilot during the learning process. However, the blurred line between support and substitution has created a paradox: are these tools enhancing learning or replacing it? Understanding this relevance is crucial not only for educators and policymakers but also for students aiming to prepare for real-world challenges.

Previous Research and Existing Gaps:

Several studies have examined the impact of AI in education, particularly in enhancing personalized learning experiences and streamlining administrative tasks. However, limited research focuses specifically on the behavioural patterns of AI usage among students and how these behaviours affect core academic values like honesty, effort, and independent problem-solving.

Additionally, while performance metrics such as GPA have been analysed in relation to AI-facilitated learning, few studies holistically address how such tools influence students' long-term preparedness for the workplace — an area beyond grades. Moreover, the demographic influences (e.g., age, gender, education level) on AI adoption behaviours in academic contexts remain underexplored. This research aims to bridge these gaps.

Research Questions:

- How does the use of AI tools affect academic integrity in terms of ethical standards, honesty in assessments, and assignment submissions?

- Does the frequency of AI use correlate with improved or diminished academic performance?
- Can AI-assisted learning help students develop skills relevant to real-world workplace environments?
- What are the common behavioural patterns and preferences among students regarding AI tool usage?
- Do demographic variables influence students' perception and usage of AI in academic settings?

Hypothesis Statement:

“AI Tools and Student Performance: Understanding the impact of AI adoption on academic integrity and future workplace readiness.”

Objectives of the Research

Stage 02 in the research

Objectives of the research:

Main Objective:

The primary objective of this research is to investigate the role of AI tools in shaping student performance, with a particular focus on their influence on academic integrity and future workplace readiness.

Specific Sub-Objectives:

To achieve the main goal, this study is guided by the following sub-objectives:

- To assess how the adoption of AI tools impacts academic integrity — particularly in relation to ethical concerns, honesty during assessments, and assignment submission practices.
- To examine whether frequent AI usage enhances or diminishes academic performance, measured through students' GPA and self-reported learning efforts.
- To explore the influence of AI usage on students' preparedness for real-world, workplace-level problem-solving beyond academic achievement.
- To analyse AI usage behaviour — including frequency, purposes, and preferred tools — to develop meaningful usage profiles.

To investigate the influence of demographic factors such as age, gender, and education level on students' academic use and perception of AI technologies.

Data Collection & Methodology

Stage 03 in the research

Data Collection & Methodology:

Population and Sample:

The target population for this study comprises Fast National University, Lahore Campus students enrolled in undergraduate and graduate programs. A convenience sampling approach was employed, and data were collected from a total of **50 respondents**. These participants varied in age, education level, and gender, providing a diverse sample to analyse AI tool usage patterns and perceptions across different demographics.

Data Collection Method:

Primary data were collected using an online survey administered through **Google Forms**. The survey was designed to ensure ease of access and response, allowing for broader participation. The form consisted of **15 structured questions**, covering multiple dimensions of AI usage, including frequency, purpose, perceived impact, and ethical perspectives.

Variables & Survey Design:

The questionnaire included both **categorical and ordinal variables**, capturing information across the following domains:

- **Demographics:** Age group, gender, and education level.
- **AI Usage Patterns:** Frequency of use, academic tasks where AI is applied, preferred tools
- **Perceptions:** Ethical concerns, honesty in quizzes/exams, guilt, academic performance (GPA), and self-perceived learning impact
- **Behavioural Attitudes:** Whether the student would perform the same without AI, perception of AI restriction, and disclosure.
- **Preparedness:** Influence of AI on real-world or workplace readiness

All variables were carefully classified into **nominal** or **ordinal** measurement scales, depending on their nature, and were later analysed using descriptive and inferential statistics.

Ethical Considerations:

To ensure ethical compliance, the survey included a short preamble informing respondents that participation was voluntary, and responses would remain anonymous and confidential. No personally identifiable information was collected. The data collected were used strictly for academic research purposes, in alignment with basic ethical research standards.

Data Analysis & Interpretation

Stage 04 in the research

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Summary of the collected data and their outcomes:

Tabular Summary:

The whole collected data was summarized in the tabular form with the frequencies of every response against each variable in the *project excel file*.

Some of the major Tabular Summaries are as follow:

Age_Group	Frequency	Relative Frequency	%age
Under 18	0	0	0
18-21	31	0.62	62
22-25	14	0.28	28
26 or above	5	0.1	10
Total	50	1	100

Education Level	Frequency	Relative Frequency	%age
Undergraduate	38	0.76	76
Graduate	7	0.14	14
Postgraduate	4	0.08	8
Others	1	0.02	2
Total	50	1	100

AI Usage Attribute	Frequency	Relative Frequency	Percentage
Almost Always	17	0.34	34
Frequently	15	0.3	30
Occasionally	13	0.26	26
Rarely	5	0.1	10
Never	0	0	0
Total	50	1	100

Most used AI	Frequency	Relative Frequency	%age
Chat GPT	43	0.86	86
Google Gemini	2	0.04	4
Apple Intelligence	1	0.02	2
Meta	4	0.08	8
Google AI Studio	0	0	0
Google NoteBook LM	0	0	0
Total	50	1	100

AI Usage improves work	Frequency	Relative Frequency	%age
Strongly Agree	16	0.32	32
Agree	25	0.5	50
Neutral	7	0.14	14
Disagree	2	0.04	4
Strongly Disagree	0	0	0
Total	50	1	100

Can perform same without using AI?	Frequency	Relative Frequency	%age
Yes, Definitely	12	0.24	24
May be with more effort	30	0.6	60
No, I rely on them	7	0.14	14
Not sure	1	0.02	2
Total	50	1	100

Interpretation of Tabular Summary:

The above Tabular Summaries indicate that:

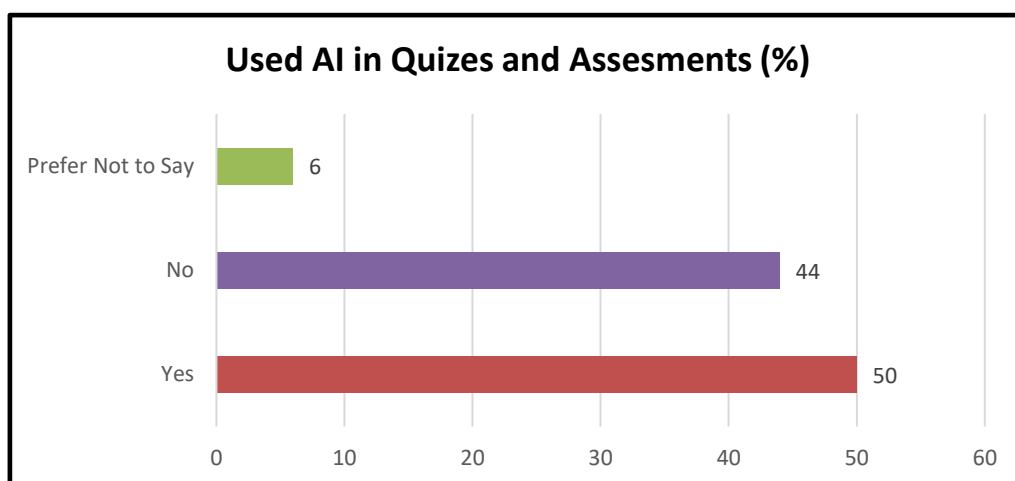
Most of the volunteers, who participated in our survey are from the age group of 18-21 making the 62% of our sample size and have an education level of undergraduate.

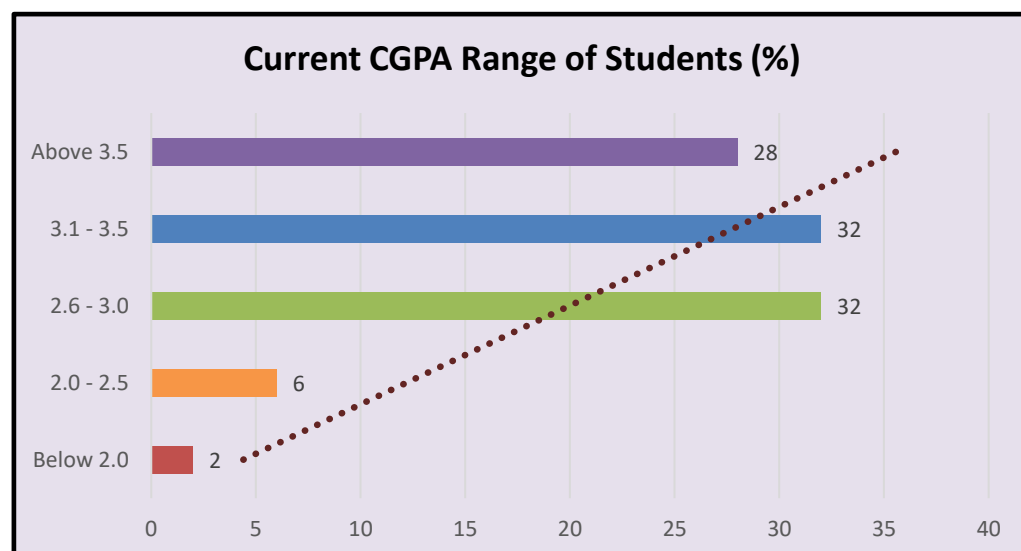
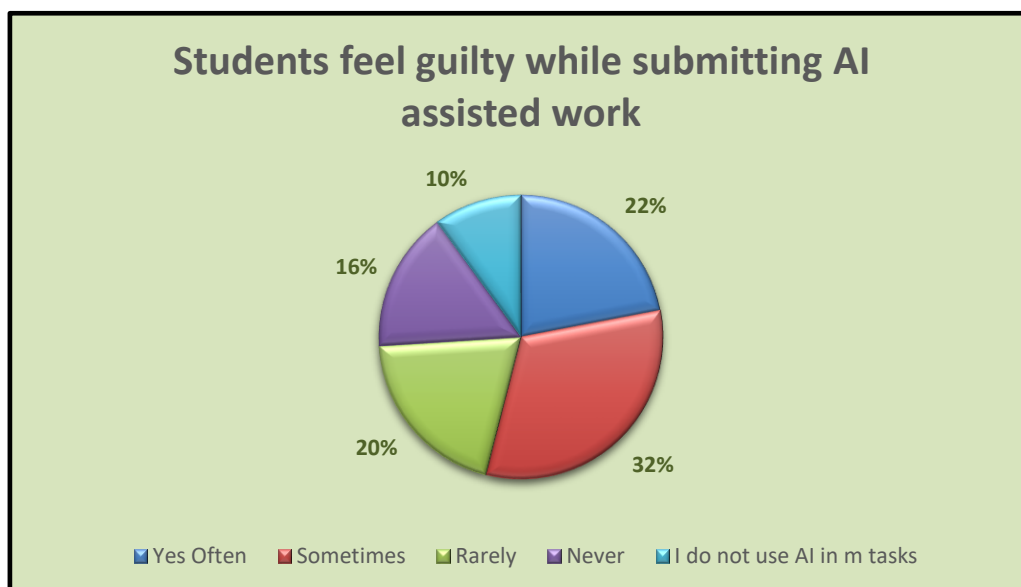
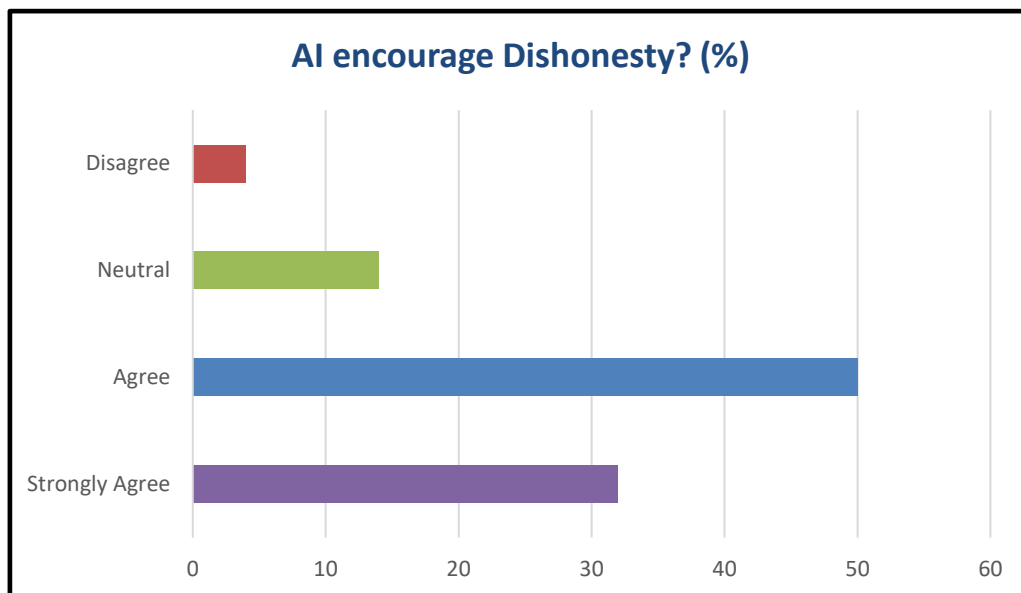
They use AI tools almost always and GPT is the top priority tool for them right now with the significant figure of 86% and almost 82% of the sample size agree as well as strongly agree that Frequently AI Usage has increase their quality of academic work.

In the shocking significant event, when the volunteers were asked can they perform same without using AI? So, almost 60% of the volunteers said they might need more effort for that, which was leading to the speculation that “their ability to think critical and in analytical was as a student is reducing?”

Graphical Summaries:

As like the tabular summaries were interpreted and summarized, we will now look forward to the graphical/visual summaries of the data:





Interpretation of Graphical Summaries:

The above graphical summaries indicate that:

In the shocking turn of the events, 50% of the students admitted that they used AI in their assessments and quizzes through unethical means and 20% of them believe that these unethical ways do not encourage any dishonesty, while 30% of them using unfair means still have some belief that these unethical ways are totally the acts of dishonesty.

When they were asked about whether they feel guilty or not after submitting the AI plage work, so surprisingly 52% of the students feel guilty often or sometimes. But 36% of them rarely or never feel guilty because they have linked these unethical ways of using integrated AI in assessments to something bigger, which they should have get through critical thinking and hard work, but they are relying on unethical means to get the timely advantage, but want to destroy their future.

And that big thing is CGPA range, almost 32% of students admitted that they have boosted their CGPA through using unethical means to get ahead in race, but at the same time they also admitted that their too much reliance on AI tools usage has slightly as well as significantly decreased their ability to think critically by a major blow of 62%.

Specific Objectives Interpretation and their outcomes:

The Leading Paradox:

The above initial shocking findings and speculations led us to formulate our hypothesis and then to creating our major objectives which we analyzed and addressed deeply in *our project excel file*.

Objective: Academic Ethical Integrity & AI tools adoption:

In this objective we analyzed the behaviors of both students' samples who used AI in their assessments and who did not use. The detailed working, analysis and charts have been linked to excel file of project. Please refer to the file.

Interpretation of Objectives sheet:

*We got the probability **P (Dishonest & Unethical Means)** as **0.50** which is making 50% chances that the students will repeat the mistakes again in the future the same way they have been using unethical means for the timely advantage in the form of CGPA booster.*

When we collectively compared students of both categories in our chart 03, we realized that almost 70% of the students were more in the weightage than those who still believe on ethical ways of getting success.

This means the institutions and scholars must shed the light on these serious issues and highlight the importance of the ability of critical thinking in the modern world, which again mentioned by the 62% of the student samples that they noticed the reduction in this ability since they have started relying on AI.

Other 2 major objectives:

Our remaining two major objectives have been addressed, analyzed and visualized on excel sheet clearly, which are:

- *Relation between AI Usage and the academic performance.*
 - *The impact of AI Usage on critical thinking and the readiness for future workplace*
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Conclusions & Takeaways

Stage 05 in the research

Conclusion:

Key Takeaways:

This study explored the growing intersection of **AI tools and student academic life**, revealing complex patterns of usage, perception, and impact. The findings suggest that AI tools are not only widely adopted among students but also deeply influence academic behaviour, performance, and future readiness. However, ethical concerns such as honesty during assessments and the tendency to withhold disclosure were also prevalent, hinting at a potential trade-off between convenience and integrity.

How Objectives were met?

Each of the research objectives was addressed through a systematic and statistically sound approach:

- **Academic Integrity:** Variables related to guilt, honesty, and disclosure were analysed, uncovering that while most students use AI, many hesitate to admit it — a key indicator of ethical tension.
- **Academic Performance:** Self-reported effort and GPA distributions were examined against AI usage frequency. While some students reported enhanced output, others admitted to a reduced sense of learning, raising questions about the true efficacy of AI in improving performance.
- **Workplace Preparedness:** Responses indicated mixed opinions — some students felt more empowered to solve real-world problems, while others questioned their independence without AI assistance.
- **Usage Behaviour:** Frequency, tools used, and use-cases were quantified and visualized. ChatGPT emerged as the most commonly used tool, mostly for summaries, assignments, and idea generation.
- **Demographic Differences:** Cross-tabulations and correlation tests showed some influence of education level on ethical perceptions and usage intensity, but gender and age showed minimal correlation.

Summaries of Main Insights:

- AI tools are now a mainstream support system in academia.
- There is an ethical grey area: students benefit from AI but struggle with transparency.
- AI usage impacts performance in nuanced ways — boosting convenience but risking deep learning.
- Usage patterns are shaped more by academic need than by demographic identity.
- Students are still navigating how to balance AI use with integrity and personal growth.

- In conclusion, AI's integration into education is irreversible — but it requires conscious regulation, awareness-building, and academic policy evolution to ensure its positive potential outweighs its risks.
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References & Citations

APA & MLA STYLE

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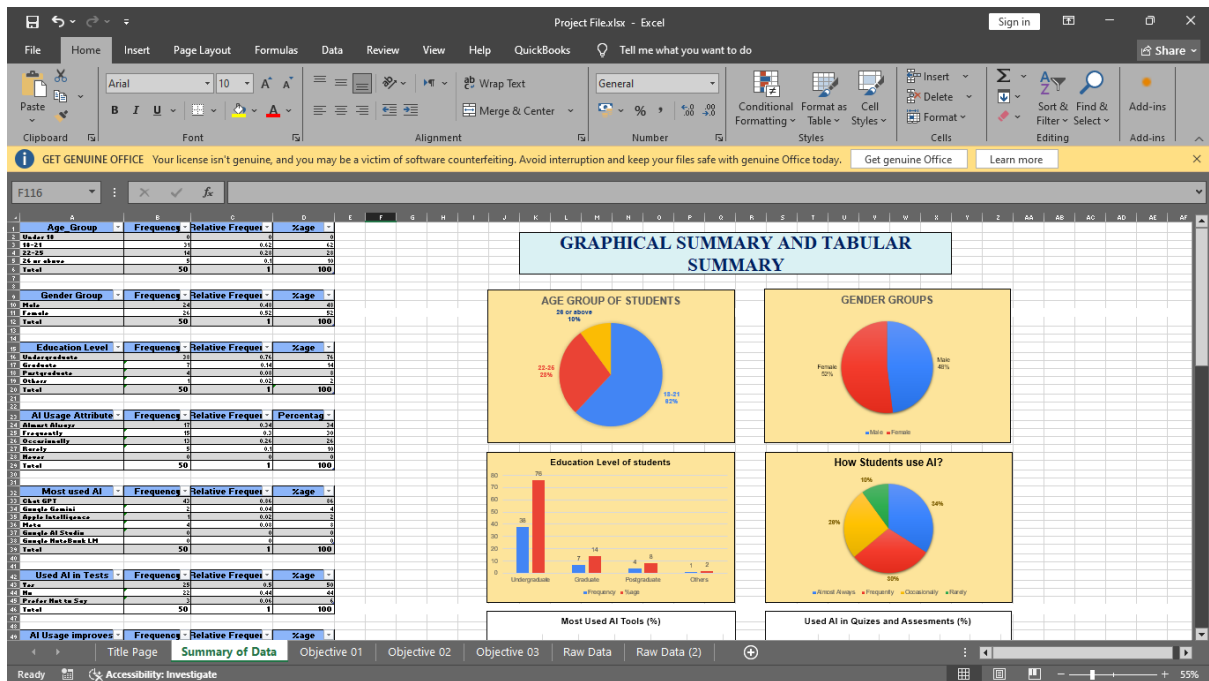
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Appendix

Working evidence



DATA SAMPLING FOR PROJECT

Hey folks, need your attention here... I am right now working on the controversial and fresh project for the course DATA ANALYSIS.. For this regard, I am collecting samples from the current students and educators.

we assure you your privacy and your answers will be anonymous..

READ THE BRIEF HERE IN IMAGE

Only serious people fill out this form..

AI Tools and Academic Integrity: Are Students relying too much on AI?

Introduction:
This survey is part of a project for the course MG2008 - Data Analysis for Business I. The data is being collected ethically by a group of 5 students from the Bachelor of Business Analytics program at FAST National University, Lahore Campus.

Purpose:
We aim to explore how frequently students use AI tools in their academic work, for which tasks, and how this usage correlates with academic performance and integrity.

Confidentiality & Ethical Assurance:

- Your responses will remain strictly confidential and will be used solely for academic and educational purposes.
- We do not collect names, roll numbers, or any personally identifying information.
- All data will be analyzed in aggregate, and no individual responses will be shared.
- By filling out this form, you consent to participate voluntarily in this survey!

Thank you for

PROJECT SURVEY

*Indicates required questions